

Assignment1: CancerMortality

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Contents

Introduction	2
First Steps	3
Explanatory Data Analysis	5
Univariate Descriptive Analysis	5
Data Quality Report	16
Imputation	18
Profiling & Relationships	20
Modeling	21
Initial Model exploration and Multicollinearity study	21
Model transformations	22
Influential data study	22
Further model exploration	22
Model validation	23
Model evaluation	23

Introduction

Please refer to the appendix at the end of the report in order to see all the code and outputs produced during the EDA. The first section contains exclusively the explanations and conclusions extracted from the outputs and plots.

This report documents the process and results of the Modeling Assignment for the Statistical Modeling and Inference course. The primary aim of this project is to analyze the Cancer Mortality dataset and build an accurate and interpretable linear model to predict `target_deathrate`, which represents the average cancer mortality rate per 100,000 individuals.

The project begins with data preparation and exploratory analysis, focusing on cleaning and enhancing the dataset to ensure its quality. This involves generating a Data Quality Report, identifying and handling missing values through imputation, and analyzing variable distributions and relationships. Univariate descriptive analysis is performed to examine each variable's characteristics.

The second stage involves modeling using the training dataset, where several key analyses are conducted to optimize the model:

- Initial model exploration addresses potential issues such as multicollinearity and the contribution of each predictor.
- Model transformations are applied to improve variable relationships and address non-linearity.
- An influential data study identifies and evaluates the impact of unusual observations on the model.
- The model is further refined through validation and diagnostics, ensuring it adheres to key assumptions like normality, homoscedasticity, and independence.

Finally, the evaluation process assesses the model's generalization performance on the test dataset, providing insights into its predictive capabilities. A detailed interpretation of the model results is presented, highlighting how the predictors influence the target variable and their practical implications for understanding cancer mortality trends.

This report follows a systematic and methodical approach to ensure the development of a reliable model.

First Steps

To begin, we import the necessary data, load the required libraries, and create functions to streamline repetitive processes.

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  Boxplot(variable, main = "Boxplot")
  outliers <- boxplot.stats(data)$out

  varsumm <- summary(data)
  Q1 <- varsumm[2]
  Q3 <- varsumm[5]
  IQR <- Q3 - Q1

  lower_severe_bound <- Q1 - 3 * IQR
  upper_severe_bound <- Q3 + 3 * IQR
  lower_mild_bound <- Q1 - 1.5 * IQR
  upper_mild_bound <- Q3 + 1.5 * IQR

  # Identify outliers
  mild_outliers <- which((data < lower_mild_bound) | (data > upper_mild_bound))
  severe_outliers <- which((data < lower_severe_bound) | (data > upper_severe_bound))
  num_mild_outliers <- length(mild_outliers)
  num_severe_outliers <- length(severe_outliers)

  # Return the results as a list
  return(list(
    num_outliers = num_mild_outliers,
    num_severe_outliers = num_severe_outliers,
    outliers_indexes = mild_outliers
  ))
}

summary_and_normality_check <- function(data) {
  #Basic summary
  print("Summary: ")
  print(summary(data))

  # Plot normality
  print("Mean")
  mm <- mean(data, na.rm = TRUE); print(mm)
  print("Standard deviation")
  ss <- sd(data, na.rm = TRUE); print(ss)
  hist(data, freq = F, 100, main = "Histogram")
  curve(dnorm(x, mean = mm, sd = ss), col="red", lwd = 2, add=T)

  #Compute skewness
  print("Skewness")
  print(skewness(data, na.rm = TRUE))

  #Number of outliers
  out.res <- calculate_outliers(data)
```

```

print("Num Outliers:")
print(out.res$num_outliers)
print("Num. severe outliers:")
print(out.res$num_severe_outliers)

#Shapiro test
print("Shapiro test: ")
print(shapiro.test(data))

# Most repeated value:
freq_table <- table(variable)
most_repeated_value <- names(freq_table)[which.max(freq_table)]; most_repeated_value
most_repeated_count <- max(freq_table); most_repeated_count
print("Most repeated value ad times it occurs: ")
print(most_repeated_value)
print(most_repeated_count)

return(out.res$outliers_indexes)
}

outliers_df <- df
outliers_df[,] <- 0

```

Explanatory Data Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the dataset, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test dataset.

Univariate Descriptive Analysis

In this initial phase, we analyze all variables in the dataset, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones.

Subsequently, we present a data quality report that assesses the overall quality of the data and discusses how it may affect the modeling process.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” First, note that it contains 126 observations with the value “1962.667684”, which we consider inconsistent and that represent NAs. After setting this individuals as NAs, we start the variable analysis.

The difference between the mean (524.24) and the median (155) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (74 and 383, respectively) are much lower than the mean and the maximum value of 38150, further suggesting skewness. As expected, the Shapiro-Wilk normality test gets rejected. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 207 mild outliers and 131 severe outliers.

```
variable <- df$avganncount
l11 <- which(variable == 1962.667684 ); length(l11)
variable[variable == 1962.667684 ] <- NA
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"avganncount"] <- 1
outliers <- which(outliers_df$avganncount == 1); length(outliers)
```

Variable *avgdeathsperyear*

avgdeathsperyear is a numerical variable representing the “Average number of reported cancer-related mortalities per year.”

This variable is complete, with no missing values. Descriptive statistics reveal a mean of 191.6 and a median of 62, indicating a strongly non-symmetric distribution. This asymmetry is further supported by the first quartile (29) and third quartile (140.5), both significantly lower than the mean, suggesting a right-skewed distribution. Visual exploration using a boxplot confirms this skewness, displaying a pronounced right tail.

Additionally, a histogram with an overlaid normal curve underscores the substantial deviation from normality, a conclusion reinforced by the results of the Shapiro-Wilk test. The distribution exhibits a significant number of outliers, with 225 classified as mild and 138 as severe, all of which are above the third quartile. These

outliers contribute to the variable's extended right tail and may warrant further investigation to understand their context and potential implications.

```
variable <- df$avgdeathsperyear
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "avgdeathsperyear"] <- 1
```

Variable *target_deathrate* :

target_deathrate is the numeric target variable of the dataset. While a more detailed analysis will be presented in subsequent sections, we introduce key insights regarding its distribution, missing values, and outliers here.

The target variable is complete in the training sample, with no missing values. The median (178.3) and mean (178.8) are nearly identical, suggesting a symmetric distribution. This symmetry is further supported by visual inspections of the boxplot and histogram, both of which confirm the balanced nature of the data around its center.

To evaluate normality, we performed the Shapiro-Wilk test. The resulting small p-value rejected the null hypothesis, indicating that the distribution deviates from normality despite its apparent symmetry. Thus, the target variable is not normally distributed.

In terms of outliers, the variable contains 35 observations classified as outliers, including one severe case.

```
variable <- df$target_deathrate;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "target_deathrate"] <- 1
```

Variable *incidencerate* :

incidencerate is a numerical variable representing the “Mean per capita cancer diagnosis rate per 100,000 individuals.”

Anomalies were identified in the data: the same 126 individuals who had a value of “1962.667684” in the *avganncount* variable also showed a value of “453.5494221” in the *incidencerate* variable. These values are considered non-coherent, and thus, we set them as missing (NA). After setting these values to NA, we proceed with the analysis of the column's distribution.

The mean of the variable is 448.7, the median is 448.7, and the standard deviation is 57.85, indicating moderate variability in the data. The skewness of 1.40 suggests a positively skewed distribution, with the tail extending toward higher values. This is further confirmed by the presence of 36 outliers, including 4 severe outliers, which are all above the upper threshold. The Shapiro-Wilk strongly rejects the null hypothesis of normality.

These results suggest that while the variable is mostly symmetric within its central range, the skewness and presence of outliers may have implications for modeling and analysis.

```
variable <- df$incidencerate;
l11 <- which(variable == 453.5494221 ); length(l11)
variable[variable == 453.5494221 ] <- NA
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "incidencerate"] <- 1
```

Variable *medincome* :

medincome is a numerical variable representing the “Median income per county.” The distribution is neither normal nor symmetric, as evidenced by the histogram, Shapiro-Wilk test, and skewness value (1.47), all indicating a right-skewed distribution with a long tail. Descriptive statistics show a minimum of 22,640, a maximum of 122,641 (Loudoun County, Virginia), and a mean of 47,278 with a standard deviation of 12,342. The median income is 45,454, with the first and third quartiles at 39,031 and 52,612, respectively. There are no missing values, but the variable contains 69 outliers, 15 of which are severe, all above the upper quartile. These findings suggest substantial variability and a pronounced concentration of counties with lower incomes compared to a smaller group with exceptionally high incomes.

```
variable <- df$medincome;  
res.var <- summary_and_normality_check(variable)  
outliers_df[res.var, "medincome"] <- 1
```

Variable *popest2015* :

popest2015 is a numerical variable representing the population of each county in 2015.

It contains no missing values but has an extensive range and an extremely skewed distribution, as highlighted by a skewness value of 14.71 and a Shapiro-Wilk test confirming strong deviation from normality. The distribution is heavily right-skewed, with a mean of 106,841 significantly exceeding the median of 27,158, and a wide spread between the 1st quartile (12,191) and 3rd quartile (66,880). The variable includes 252 mild outliers and 162 severe outliers, all exceeding the upper whisker of the boxplot. The highest value, 10,170,292, corresponds to Los Angeles County, California. The standard deviation of 376,494 reflects the substantial variability, driven by a small number of extremely populous counties. This skewed distribution highlights the vast disparity in county populations, with most counties having smaller populations and a few extreme outliers significantly influencing the overall statistics.

```
variable <- df$popest2015;  
res.var <- summary_and_normality_check(variable)  
outliers_df[res.var, "popest2015"] <- 1
```

Variable *povertypercent* :

povertypercent is a numerical variable representing the percentage of the population in poverty. It has no missing values and a moderate positive skew, as indicated by the skewness value of 0.89 and the Shapiro-Wilk test, confirming the variable does not follow a normal distribution. The mean is 16.79, slightly higher than the median of 15.7, reflecting the slight skew. The 1st quartile (12.15) and 3rd quartile (20.4) show a moderate spread, with values ranging from a minimum of 3.7 to a maximum of 44. The standard deviation is 6.41, indicating variability in poverty levels across counties.

The variable has 42 mild outliers, but no severe ones. The most frequent value is 14.7, occurring 22 times. While this variable shares a non-normal distribution with other numerical variables like *medincome* and *incidencerate*, its skewness and outlier characteristics are less pronounced, indicating a more balanced distribution overall.

```
variable <- df$povertypercent;  
res.var <- summary_and_normality_check(variable)  
outliers_df[res.var, "povertypercent"] <- 1
```

Variable *studypercap* :

studypercap is a numerical variable representing the “Per capita number of cancer-related clinical trials per county.”

A significant feature of this variable is that 1,162 out of 1,831 counties (63.5%) have a value of 0, making this the most frequent value and dominating the distribution. Nearly all non-zero values are flagged as outliers due to their extreme deviation from the majority, with 307 identified as mild outliers, and 222 severe ones.

Given the predominance of zero values and the highly skewed nature, this variable may contribute more noise than predictive value to the model. **Removing it from both the training and test datasets is a reasonable decision to simplify the model and enhance its performance.**

```
variable <- df$studypercap;
res.var <- summary_and_normality_check(variable)

# Delete from train, test
lltrain <- which(names(df) == "studypercap");lltrain
lltest <- which(names(test) == "studypercap");lltest
df <- df[,-lltrain]
test <- test[,-lltest]
```

Variable *binnedi* :

The variable *binnedi* represents median income per capita binned into deciles, categorizing counties into ten groups based on their income distribution. The frequency table shows a relatively balanced distribution across these bins, with counts ranging from 164 to 195 observations per bin. For instance, the smallest bin (37413.8, 40362.7] contains 164 observations, while the largest bin (48021.6, 51046.4] contains 195. The proportion table confirms this balance, with percentages varying narrowly from 8.96% to 10.65%. It contains no missing values.

Given its nature, *binnedi* is likely related to *medincome*, the median income per county, as both variables represent different perspectives of income distribution. While *medincome* is a continuous variable, *binnedi* simplifies analysis by discretizing income into clear categories. This potential relationship highlights the need to evaluate *binnedi* alongside *medincome* to assess redundancy or complementarity in modeling efforts.

```
variable <- df$binnedinc;
summary <- summary(variable); summary
table(variable)
prop.table(table(variable))
barplot(table(variable), main="Bar Plot of Variable", xlab="Categories", ylab="Frequency")
```

Variable *medianage* :

medianage is a numerical variable that expresses the median age of county residents.

Note that initially, the variable contained 18 nonsensical values, ranging from 349.2 to 624, which were considered incoherent. These values were replaced with NAs, leaving the variable with no other missing values. The cleaned data shows a median age ranging from 23.3 to 59.0 years, with a mean of 40.85 and a standard deviation of 5.14. The skewness is near zero (-0.024), indicating a nearly symmetric distribution. However, the Boxplot and Shapiro-Wilk test for normality suggests the variable is not normally distributed despite its symmetry. The data contains 52 mild outliers (both above the upper bound and below the lower bound), but none are classified as severe. The most frequent value is 38.8, which occurs 29 times.

```
variable <- df$medianage
table(variable)
df[which(df$medianage > 60), c("medianage")] <- NA
summary(df$medianage)
res.var <- summary_and_normality_check(df$medianage)
outliers_df[res.var, "medianage"] <- 1
```

Variable *medianagemale* :

medianagemale represents the median age of male county residents.

All its values are reasonable, ranging from 23.0 to 60.2 years, with no incoherent or missing entries. The variable has a mean of 39.59 years, a median of 39.50 years, and a standard deviation of 5.19, indicating a central tendency and variability. The skewness is slightly positive (0.161), suggesting a slight asymmetry with a longer right tail. However, the Shapiro-Wilk normality test indicates the distribution deviates significantly from normality. The dataset contains 46 mild outliers.

Comparing *medianagemale* with *medianage*, the two variables exhibit similar distributions and characteristics, which is expected as *medianage* includes both genders.

```
variable <- df$medianagemale
res.var <- summary_and_normality_check(variable)
table(variable)
outliers_df[res.var, "medianagemale"] <- 1
```

Variable *medianagefemale* :

medianagefemale represents the median age of female county residents.

Its values range from 23.6 to 58.2 years. All entries are valid, showing no nonsensical or missing values. The average median age is 42.17 years, slightly higher than that of males, with a median of 42.40 years and a standard deviation of 5.20 years. The data is slightly skewed to the left, as indicated by a skewness of -0.206, and the Shapiro-Wilk test confirms a significant deviation from normality. There are 55 mild outliers.

```
variable <- df$medianagefemale
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "medianagefemale"] <- 1
```

Variable *geography* :

The variable *geography* is a categorical variable that specifies the name of the county for each observation. It contains no missing values and serves as a unique identifier for each row, as all values are distinct. Additionally, it includes the name of the state to which each county belongs. To potentially enhance the model, we propose extracting the state names from *geography* and creating a new factor representing states. This new factor could capture regional effects and improve the model's ability to identify patterns related to state-level characteristics.

```
variable <- df$geography;
table(variable)

#Create new factor (in the train and test set)
```

```
df$state <- sapply(strsplit(df$geography, ", "), function(x) x[2])
test$state <- sapply(strsplit(test$geography, ", "), function(x) x[2])
table(df$state)
```

Variable *percentmarried*

The variable *percentmarried* represents the percentage of county residents who are married.

It contains no errors, as all values fall within the valid range of 0 to 100. The variable has a minimum of 23.1% and a maximum of 68.0%, with a mean of 51.9% and a median of 52.5%. The standard deviation is 6.74%, indicating moderate variability in marriage rates across counties. The distribution is slightly left-skewed, with a skewness of -0.65, suggesting a higher concentration of values near the upper end of the range. Although the Shapiro-Wilk test confirms that the distribution is not normal, this skewness does not indicate extreme asymmetry. There are 34 outliers, but none are classified as severe. The most frequent value is 52.2%, which appears 22 times.

```
variable <- df$percentmarried
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "percentmarried"] <- 1
```

Variable *pctnohs18_24*

The numerical variable *pctnohs18_24* represents the percentage of county residents aged 18-24 whose highest level of education is less than high school. The values range from 0.5% to 59.1%, with a mean of 18.29% and a median of 17.2%. The distribution of the variable is positively skewed, with a skewness of 0.89, suggesting that a higher proportion of counties have lower values (fewer residents with less than high school education), while some counties exhibit higher percentages. The Shapiro-Wilk test for normality confirms a significant deviation from normality. There are 35 outliers, with 5 classified as severe.

```
variable <- df$pctnohs18_24
table(df$pctnohs18_24)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pctnohs18_24"] <- 1
```

Variable *ppcths18_24*

The variable *ppcths18_24* represents the percentage of county residents aged 18-24 whose highest level of education is a high school diploma.

Note that there is a county with a 0% (Greeley County, Kansas), which potentially is an error.

The values range from 0% to 72.5%, with a mean of 35% and a median of 34.7%. The standard deviation is 8.96%, indicating substantial variability in high school graduation rates across counties. The skewness is 0.22, suggesting a slight positive skew, meaning that most counties have lower percentages of residents with only a high school diploma, while a few counties exhibit higher values. The Shapiro-Wilk test confirms that the distribution is not normal. There are 32 outliers, none of which are severe.

This variable is related to *pctnohs18_24* (the percentage of residents aged 18-24 without a high school diploma), and we can consider creating a factor variable for characterizing this relation.

```
variable <- df$pcths18_24
ll<- which(df$pcths18_24 == 0);
res.var <- summary_and_normality_check(df$pcths18_24)
outliers_df[res.var,"pcths18_24"] <- 1
table(df$pcths18_24)
```

Variable *pctsomecol18_24*

The variable *pctsomecol18_24* represents the percentage of county residents aged 18-24 whose highest level of education attained is some college. However, this variable contains 1,376 missing values, which accounts for approximately 75% of the total observations. Given the high proportion of missing data, further steps will be taken to address the missing values, either through imputation methods to obtain valid results or by deleting the variable if it cannot be reasonably imputed.

```
summary(df$pctsomecol18_24)
```

Variable *pctbachdeg18_24*

The variable *pctbachdeg18_24* represents the percentage of county residents aged 18-24 who have attained a bachelor's degree. It ranges from 0% to 51.8%, with a mean of 6.22% and a median of 5.4%. The standard deviation is 4.52%, indicating notable variability across counties. A key feature of this variable is the presence of 73 zero values, meaning that in many counties, no residents aged 18-24 have a bachelor's degree. The skewness is 1.98, reflecting a strongly positive skew, where most counties have relatively low percentages, while a few counties exhibit exceptionally high values. The Shapiro-Wilk test indicates that the normality assumption is rejected. The variable contains 56 mild outliers, and 10 of them classified as severe.

It may also be particularly insightful when combined with related variables, such as *pcths18_24* and *pctsomecol18_24*, to explore the full spectrum of educational attainment.

```
variable <- df$pctbachdeg18_24
table(df$pctbachdeg18_24)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctbachdeg18_24"] <- 1
```

Variable *pcths25_over*

The variable *pcths25_over* represents the percentage of county residents aged 25 and over whose highest level of education is a high school diploma.

It ranges from 8.3% to 52.7%, with a mean of 34.73% and a median of 35.3%. The standard deviation is 7.07%. The skewness of -0.37 reflects a slight left skew, suggesting a concentration of values toward the higher end of the range. The Shapiro-Wilk test indicates that the normality assumption is rejected. The variable has 18 outliers, none of which are severe. The most frequent value is 35.6%, occurring 19 times.

```
variable <- df$pcths25_over
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pcths25_over"] <- 1
```

Variable *pctbachdeg25_over*

The variable *pctbachdeg25_over* captures the percentage of county residents aged 25 and over with a bachelor's degree as their highest level of education.

The distribution is positively skewed, with most counties at the lower end of the range (2.5% to 42.2%), while a few have much higher values. The mean is 13.3%, close to the median of 12.3%, with a standard deviation of 5.47%. Outliers are identified in 59 counties, including 3 severe ones. The variable does not follow a normal distribution.

This variable is essential for understanding higher education attainment in adults and its variability across counties. When analyzed alongside related variables like *pcths25_over*, it can help reveal regional educational trends.

```
table(df$pctbachdeg25_over)
res.var <- summary_and_normality_check(df$pctbachdeg25_over)
outliers_df[res.var, "pctbachdeg25_over"] <- 1
```

Variable *pctemployed16_over*

pctemployed16_over represents the percentage of county residents aged 16 and over who are employed. It ranges from 23.9% to 80.1%, with a mean of 54.21% and a median of 54.5%. The data has a standard deviation of 8.22%. Its skewness of -0.34 suggests a slight left skew, meaning there are slightly more counties with higher employment percentages. The Shapiro-Wilk test rejects the normality assumption. There are 11 outliers, none of which are severe.

Additionally, 82 missing values (NA) are present, which require further handling (imputation section).

```
table(df$pctemployed16_over)
res.var <- summary_and_normality_check(df$pctemployed16_over)
outliers_df[res.var, "pctemployed16_over"] <- 1
```

Variable *pctunemployed16_over*

The variable *pctunemployed16_over* represents the percentage of county residents aged 16 and over who are unemployed. It spans from a minimum of 0.7% to a maximum of 29.4%, with a mean of 7.86% and a median of 7.5%. The standard deviation is 3.46%, indicating moderate variability across counties. The distribution is positively skewed, with a skewness of 0.92, suggesting that while most counties have lower unemployment rates, a few exhibit significantly higher percentages. The Shapiro-Wilk test confirms that the normality assumption is rejected. There are 42 outliers, including 4 severe ones.

This variable, in conjunction with *pctemployed16_over*, provides a comprehensive view of the labor market across counties.

```
table(df$pctunemployed16_over)
res.var <- summary_and_normality_check(df$pctunemployed16_over)
outliers_df[res.var, "pctunemployed16_over"] <- 1
```

Variable *pctprivatecoverage*

pctprivatecoverage is a numerical variable that reflects the percentage of county residents with private health coverage. The values span from 23.4% to 89.6%, with a median of 65.2% and a mean of 64.47%. The interquartile range is relatively narrow, from 57.5% (1st quartile) to 72.1% (3rd quartile), indicating that most

counties have moderate to high levels of private health coverage. The standard deviation is 10.64%, showing some variability, while the skewness of -0.38 points to a slight left skew, with more counties concentrated at higher percentages. The distribution rejects the normality assumption, as confirmed by the Shapiro-Wilk test.

The dataset identifies 17 outliers, but none are severe. These represent counties with exceptionally low private health coverage rates, offering insights into regional disparities.

This variable holds significant importance due to its potential connection with the target variable, cancer target deathrate. Private health coverage likely plays a crucial role in access to healthcare services, early detection, and treatment of cancer, making it a key factor in understanding regional variations in cancer-related mortality.

```
table(df$pctprivatecoverage)
res.var <- summary_and_normality_check(df$pctprivatecoverage)
outliers_df[res.var, "pctprivatecoverage"] <- 1
```

Variable *pctprivatecoveragealone*

The variable *pctprivatecoveragealone* represents the percentage of county residents with private health coverage without any public assistance. It ranges from 16.8% to 78.9%, with a mean of 48.65% and a median of 49.0%, indicating that approximately half of the counties have coverage levels close to the central tendency. The interquartile range spans from 41.5% to 55.5%, with a standard deviation of 10.07%. The skewness of 0.02 indicates an almost symmetrical distribution, and the Shapiro-Wilk test suggests the data does not deviate significantly from normality. However, the variable contains 356 missing values (approximately 20% of the dataset), which may require imputation or exclusion strategies. There are 4 outliers, none of which are severe.

This variable is closely related to *pctprivatecoverage*, which captures overall private health coverage, including public assistance.

```
table(df$pctprivatecoveragealone)
res.var <- summary_and_normality_check(df$pctprivatecoveragealone)
outliers_df[res.var, "pctprivatecoveragealone"] <- 1
```

Variable *pctempprivcoverage*

pctempprivcoverage represents the percentage of county residents with employee-provided private health coverage.

Its values range from 14.3% to 70.2%, with a mean of 41.29% and a median of 41.1%, indicating a balanced central tendency. The interquartile range spans from 34.6% to 47.7%, with a standard deviation of 9.53%. The skewness is close to zero (0.11), suggesting a nearly symmetrical distribution, while the Shapiro-Wilk test indicates a slight deviation from normality for a confidence level of 99%, but is not rejected in a confidence of 95% (p-value = 0.02861). The variable contains 7 outliers, none of which are severe.

This variable is integral to understanding employment-related health benefits, especially when analyzed alongside *pctprivatecoverage* and *pctprivatecoveragealone*.

```
table(df$pctempprivcoverage)
res.var <- summary_and_normality_check(df$pctempprivcoverage)
outliers_df[res.var, "pctempprivcoverage"] <- 1
```

Variable *pctpubliccoverage*

The numeric variable *pctpubliccoverage* represents the percentage of county residents with government-provided health coverage, which is not exclusive and may overlap with other types of coverage. It ranges from 11.2% to 62.7%, with a mean of 36.15% and a median of 36.3%, suggesting a balanced distribution around the central tendency. The interquartile range extends from 30.9% to 41.4%, and the standard deviation of 7.84% . The skewness of -0.03 reflects a nearly symmetrical distribution. The Shapiro-Wilk test suggests that the data does not significantly deviate from normality. There are 13 outliers, though none are classified as severe.

```
table(df$pctpubliccoverage)
res.var <- summary_and_normality_check(df$pctpubliccoverage)
outliers_df[res.var, "pctpubliccoverage"] <- 1
```

Variable *pctpubliccoveragealone*

The variable *pctpubliccoveragealone* represents the percentage of county residents with government-provided health coverage exclusively. The data ranges from 2.6% to 46.6%, with a mean of 19.15% and a median of 18.7%. Most counties fall within an interquartile range of 14.9% to 23%, while the standard deviation of 6.06% indicates moderate variability. The skewness of 0.47 suggests a slight positive skew, with more counties having lower values. The Shapiro-Wilk test indicates that the normality assumption is rejected, and there are 21 outliers, though none are severe.

This variable should complement *pctprivatecoveragealone*, which measures private coverage without public assistance. Together, they should provide a complete view of the reliance on public versus private health coverage. Counties with higher *pctpubliccoveragealone* may reflect areas with greater socioeconomic challenges or fewer employer-provided insurance options.

```
table(df$pctpubliccoveragealone)
res.var <- summary_and_normality_check(df$pctpubliccoveragealone)
outliers_df[res.var, "pctpubliccoveragealone"] <- 1
```

Variable *pctwhite*

Numerical variable that represents the percent of county residents who identify as White.

Both from the plot and from inferential methods we can easily discard that this variable is anywhere near normal, being a hugely right skewed variable. It contains 104 outliers from which 7 are severe.

This points out that the vast majority of counties have a high percentage of white population.

– op2 : mild = 97 | sev = 7

```
res.var <- summary_and_normality_check(df$pctwhite)
outliers_df[res.var, "pctwhite"] <- 1
```

Variable *pctblack*

Numerical variable that represents the percent of county residents who identify as Black.

There is a huge number of outliers (323), from which 99 are severe and there is also a huge left skewness, that points out that the vast majority of counties (except the outliers) have a rather low percentage of black folks. The distribution almost looks like a power law distribution (Zipf) therefore we can safely reject both with visualization and inferential methods that the distribution of this variable is not normal.

– op2: mild = 224 | sev = 99

```
res.var <- summary_and_normality_check(df$pctblack)
outliers_df[res.var, "pctblack"] <- 1
```

Variable *pctasian*

Numerical variable that represents the percent of county residents who identify as Asian.

Pretty similarly to the last distribution, we have a heavily left skewed zipfian like distribution, with 305 outliers (from which 107 are severe), pointing out that the vast majority of counties have less than 3% of asian percentage.

– op2: mild = 198 | sev = 107 – A: no acabo de entendre el resultat obtingut al boxplot dsp de mirar el histograma

```
res.var <- summary_and_normality_check(df$pctasian)
hist(df$pctasian , 50 )
outliers_df[res.var, "pctasian"] <- 1
```

Variable *pctotherrace*

Numerical variable that represents the percent of county residents who identify as any other race (Pacific islanders, arábians, saiyajins, namekians...).

Again this variable, just like the last 2 presents a huge left skewness and zipfian like distribution with 283 outliers, from which 102 are severe. Again this sampled distribution is not normal.

– op2 : mild = 181 | sev = 102

```
res.var <- summary_and_normality_check(df$pctotherrace)
outliers_df[res.var, "pctotherrace"] <- 1
```

Variable *pctmarriedhouseholds*

Numerical variable that represents the percent of married households.

This variable contains 59 outliers from which 2 are severe and a noticeable left skewness. Visualization isn't that clear, but using inferential methods like shapiro test we reject the null hypothesis that this variable is normal.

op2: mild = 57 | sev = 2

```
res.var <- summary_and_normality_check(df$pctmarriedhouseholds)
outliers_df[res.var, "pctmarriedhouseholds"] <- 1
```

Variable *birthrate*

Numerical variable that represents the number of births relative to number of women in that specific county.

Finally, this last variable contains a fair amount of outliers (121) from which 17 are severe. It also has a fair amount of left skewness. Both from visualization and shapiro testing we reject that this sample variable is normal.

op2 : mild = 104 | sev = 17

```
res.var <- summary_and_normality_check(df$birthrate)
outliers_df[res.var, "birthrate"] <- 1
```

Data Quality Report

Now that all the variables have been slightly analyzed in terms of their meaning. Given that univariate outliers and NA values have already been computed and shown, only a ranking and a visual summary will be showcased to gather the main insights visually.

This quality report will be divided between variables and individuals of the dataset, and finally a multivariate outlier analysis will be performed.

Per variable

Per variable, count:

- Number of missing values.
- Number of errors (including inconsistencies).
- Number of outliers.
- Rank variables according the sum of missing values (and errors).

```
# Number of errors (including inconsistencies). TODO
md.pattern(df, rotate.names=T)
```

On one hand, looking at the pattern, we can notice that there are 362 variables with no null values, and that the vast majority of nulls (1056) are located on the pctsomecol18_24 variable with just 1 null column. Another remarkable trend is that up to 248 individuals have 2 null values in variables pctsomecol18_24 and pctprivatecoveragealone, while this other variable just has 83 pure nulls (Only this column). The last variable that contains a low amount of null values is pctemployed16_over, where 20 individuals have the last 3 mentioned variables containing nulls, and the rest of few possible NA values come from pattern permutations of these 3 variables. Therefore this dataset only has 3 possible rows containing NA, and only one of them is relevant in this aspect. Ranking:

Rank	Null values
1	pctsomecol18_24: 1375 NA
2	pctprivatecoveragealone: 356 NA
4	avganncount : 126 NA
4	incidencerate : 126 NA
3	pctemployed16_over: 82 NA

On the other hand, the top 5 ranking for variables with the most amount of mild and severe outliers is:

TO_DO -> CHECK

Rank	Mild Outliers	Severe Outliers
1	avganncount 333	avganncount 257
2	pctblack 323	studypercap 222
3	studypercap 307	popest2015 162
4	pctasian 305	avgdeathsperyear 138

Rank	Mild Outliers	Severe Outliers
5	pctotherrace 283	pctasian 107

Per individuals

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariate outliers.

There are 362 obs without NA in any variable, 1144 obs with 1 variable with a NA value, 305 obs with 2 NA values and 20 obs with 3 NA values along the 33 variables of the dataset.

```
#Number of missing values
na_counts <- rowSums(is.na(df))
table(na_counts)
```

Out of all observations, 971 do not exhibit any outliers across the variables, while 385 observations contain a single outlier in one variable. Interestingly, there is 1 observation where we identified outliers in 10 variables and another 10 observations that are outliers in 8 variables.

```
#Number of outliers per individual
outliers_counts <- rowSums(outliers_df)
table(outliers_counts)
```

Considering over 20 numerical variables that contain few correlated variables and considering a 1% significance level, 209 multivariate outliers are found in the dataset.

```
#Identify individuals considered as multivariate outliers.
df$avgdeathspereyear <- as.numeric(df$avgdeathspereyear)
df$medincome <- as.numeric(df$medincome)
df$popest2015 <- as.numeric(df$popest2015)

#Compute multivariate outliers and find their ids (24)
subset_num_vars <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
res_mout <- Moutlier(df[,subset_num_vars], quantile=0.97, plot=T)
mout_ids <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff))

#Reset graphical parameters and plot the outliers
dev.off()
plot(x=res_mout$md, y=res_mout$rd, main="Classic Mahalanobis D vs Robust Mahalanobis D (M. Outliers)",
abline(v=res_mout$cutoff, col="red")
abline(h=res_mout$cutoff, col="red")
```

We can clearly see that for the number of outliers variable, the mean is around 1.39, and the third quantile around 2.0, however, we have a high standard deviation of 1.935 pointing out at the presence of outliers in the number of outliers variable, suggesting there are several individuals with more than 2 outlier variables. This variable contains somewhat high correlation with pcths25_over, pchwhite, pctasian and other weak correlations.

The number of missing values doesn't make much sense to be analyzed after removing 2 of the main problematic columns which contained several null values.

Finally, looking at the multivariate factor profiling, we see that there is a global association between "mout" and "binnedinc", and we can see that there is a huge global association with quantitative variables pctotherrace and n_uout (Logically, the number of univariate outliers contributes to a possible multivariate outlier individual), followed by medianage looking at their eta squared values. Finally, we see looking at the mean values of the quantitative variables that in the case of an individual that is not a multivariate outlier, almost every single variable mean value is equal or close to the overall mean. However in the case of multivariate outliers we see huge discrepancies in the mean of pctasian, pctotherrace, n_out, popest2015, avgdeathspereyear and avganncount, which are way higher than the overall mean value. This may point out that the asian-otherrace populations and those with higher population or more average deaths per year, are more prone to be multivariate outliers.

```
anal_df <- df
#Binary factor if its a multivariate outlier
anal_df$mout <- 0
anal_df[mout_ids, "mout"] <- 1
anal_df$mout <- factor(anal_df$mout, labels=c("Not M.O", "Yes M.O"))
#Number of univariate outlier variables for a given observation
anal_df$n_uout <- rowSums(outliers_df)
#Number of missing values in variables for a given observation
anal_df$n_miss <- rowSums(is.na(df))

summary(anal_df[, c("mout", "n_uout", "n_miss")]); sd(anal_df$n_uout); sd(anal_df$n_miss)
num_vars <- anal_df[sapply(anal_df, is.numeric)]
corrplot(cor(num_vars))

catdes(anal_df, num.var=34, proba=0.05)
```

Imputation

pctsomecol18_24

Given the huge amount of null values, the impute PCA has not been able to replicate the original distribution of this messy variable.

```
num_vars <- df[sapply(df, is.numeric)]
miss_ids <- which(is.na(df$pctsomecol18_24))
res_imp <- imputePCA(num_vars)
imp_pctsomecol <- res_imp$completeObs[, "pctsomecol18_24"]

# Calculate the quantiles for validation
quantiles_df <- quantile(df$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctsomecol, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
     xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("df$pctsomecol18_24", "imp_pctsomecol"),
     col = c("blue", "red"), lwd = 2)
```

```

#We won't impute the values!
df[miss_ids, "pctsomecol18_24"] <- imp_pctsomecol[miss_ids]

# --- op2 ---
summary(df$pctsomecol18_24)
imp_values <- 100 - df$pctnohs18_24[miss_ids] - df$pcths18_24[miss_ids] - df$pctbachdeg18_24[miss_ids]
pct2 <- df$pctsomecol18_24
pct2[miss_ids] <- imp_values

quantiles_2 <- quantile(pct2, probs = seq(0,1,0.1), na.rm = FALSE)
summary(pct2)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")

lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2);
lines(seq(0, 1, 0.1), quantiles_2, col = "green", lwd = 2);
legend("topright", legend = c("df$pctsomecol18_24", "imp_pctsomecol", "imputation2"),
      col = c("blue", "red", "green"), lwd = 2)

# add imputed values
df$pctsomecol18_24 <- pct2

```

pctemployed16_over

pctemployed16_over only contains 70 null values and the imputePCA function has been able of almost perfectly preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

```

num_vars <- df[sapply(df, is.numeric)]
miss_ids <- which(is.na(df$pctemployed16_over))
res_imp <- imputePCA(num_vars)
imp_pctemployed16_over <- res_imp$completeObs[, "pctemployed16_over"]

# Calculate the quantiles for validation
quantiles_df <- quantile(df$pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("pctemployed16_over", "imp_pctemployed16_over"),
      col = c("blue", "red"), lwd = 2)

df[miss_ids, "pctemployed16_over"] <- imp_pctemployed16_over[miss_ids]

```

pctprivatecoveragealone

Similarly, the variable pctprivatecoveragealone is better informed despite a quite high number of null values (253) and the imputePCA has been successful to preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

```

num_vars <- df[sapply(df, is.numeric)]
miss_ids <- which(is.na(df$pctprivatecoveragealone))
res_imp <- imputePCA(num_vars)
imp_pctprivatecoveragealone <- res_imp$completeObs[, "pctprivatecoveragealone"]

# Calculate the quantiles for validation
quantiles_df <- quantile(df$pctprivatecoveragealone, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctprivatecoveragealone, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("df$pctprivatecoveragealone", "imp_pctprivatecoveragealone"),
      col = c("blue", "red"), lwd = 2)

df[miss_ids, "pctprivatecoveragealone"] <- imp_pctprivatecoveragealone[miss_ids]

```

Profiling & Relationships

After imputation and removal of null values, we can proceed with profiling the target variable and assess if any variable contains serial correlation.

TODO: explicar correlacions... (Cuidado, no cal explicar les correlacions de target_deathrate, ja estan just al apartat d'abaix!!)

```

names(df)
num_vars <- df[sapply(df, is.numeric)]
cat_vars <- df[!sapply(df, is.numeric)]
corrplot(cor(num_vars))

```

For some unexpected reason the geography categorical variable causes some unexpected problems to arise, therefore we won't use it at all. If you take a closer look at the profiling of this target variable you will realize that there is a high linkage between the target and the qualitative variable binnedinc performing a oneway anova test. If you take a look at the mean values for different values of binnedinc, you will see that those incomes which are below 40K have a higher deathrate (The lower the binned income the higher the deathrate). Conversely, the higher the binned income the lower the deathrate. Finally, taking a glance at the quantitative variables, we see that there is a medium-high positive correlation with pctpubliccoveragealone, povertypercent, pctpubliccoverage, pcths25_over and pctunemployed16_over. Looking at the inverse, we see a medium-high inverse correlation with variables medincome, pctbachdeg25_over and pctprivatecoverage and pctmarriedhouseholds.

```

subset_num <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
subset_cat <- c(9)
subset_df <- c(subset_num, subset_cat)

res_anal <- condens(df[, subset_df], num.var=3, proba=0.01)
res_anal

```

Serial correlations

It has been found previously that the target_variable doesn't follow a normal distribution at all, now its clear looking at the auto correlation function plot below that this variable contains serial correlation. To address this issue we will

```
library(stats)
acf(df$target_deathrate)

ids <- sample(1:nrow(df), nrow(df))
df <- df[ids, ]
acf(df$target_deathrate)
```

Modeling

Initial Model exploration and Multicollinearity study

Firstly, it is important to realize that just as in past sections, the geography variable should just be used as a supplementary variable and not used for modelling purposes. Two initial models are built, one with all the numeric variables, and a second one containing all the numeric variables + all categorical variables.

Both models explain roughly 51% and 52% of variance respectively (R^2), and both have multiple significant slope coefficients such as percentmarried, pctbachdeg25_over, pctemployed16_over, pctotherrace, avganncount, incidencerate, avgdeathsperyear and binnedinc level 7 among the ones with smallest p-values.

Looking at variance over inflation of the first model, we can see that there are several variables that contain a great amount of variance over inflation, pointing out the presence of multicollinearity. The main ranked variables that exceed the high VIF threshold (10) in both models:

1. avgdeathsperyear
2. pctpubliccoverage
3. popest2015
4. pctprivatecoveragealone
5. pctprivatecoverage
6. pctpubliccooveragealone
7. avganncount
8. percentmarried
9. binnedinc

Exploring the residual analysis plots of each model, a common trend is easily identified. Due to the presence of extreme outliers, both models do not meet the linear model premises at all:

1. Residuals vs Fitted plots look like residuals are random and do not follow any pattern except for a couple of extreme outliers, which distort the straight line, therefore we cannot claim linearity.
2. Q-Q Residuals suggest that the residuals almost follow a normal distribution (are random), despite a few outlier and heavier tails than usual, therefore we cannot fully claim normality.
3. Scale-Location plots show a slight curvature due to the outlier values that points out a possible small heteroscedasticity.
4. Residuals vs Leverage confirms that despite almost all points have a quite low leverage, there are a small amount of highly leveraged points that distort the model in general (Specially observation 600).

```
m1 <- lm(target_deathrate ~ avganncount + avgdeathsperyear + incidencerate + medincome + popest2015 + p

summary(m1)
vif(m1)
par(mfrow=c(2,2)) # Change the panel layout to 2 x 2
plot(m1, id.n=0)
```

```
# Mixed (Categorical + numerical)
m2 <- lm(target_deathrate ~ avganncount + avgdeathsperyear + incidencerate + medincome + popest2015 + p

summary(m2)
vif(m2)
plot(m2)
```

Following up, we will build m3, a modification of m2 removing some inflating variables (avgdeathsperyear, pctprivatecoveragealone, pctpubliccoveragealone, binnedinc, percentmarried, medianagefemale, medianage-male) which some of them are redundant. This is one approach for addressing multicollinearity in the model, another strategy could be to perform a dimensionality reduction technique like PCA. The results we get from this variable removal is a model with all the independent variables having a less than 10 VIF (However, we still have variables that are above 5 VIF like avganncount, popest2015...), yet still retaining over 50 R-squared. However, the residual analysis results are pretty much the same as in the past 2 models with minor improvements. Outliers keep distorting the model heavily. Anova nested model test confirms that is worth using the lighter model.

```
m3 <- lm(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypersent + stu
vif(m3)
summary(m3)
plot(m3)

#anova(m3, m2)
```

One further improvement for a following model would be to remove variables with high VIF and high p-value in the coefficient tests, like popest2015 or avganncount. Another possible improvement would be to retain binnedinc instead of its redundant counterpart medincome.

```
library(effects)
plot(allEffects(m3))
```

Model transformations

Possibly we need to perform some transformations...

```
#Exemple random ;)
#boxcox(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypersent + stud

#boxTidwell(target_deathrate ~ avganncount + incidencerate + medincome + popest2015 + povertypersent + .
```

Influential data study

Given that the models are being quite negatively affected by influential data, an study of the aforementioned will be performed to address this issue.

```
#Include Code
```

Further model exploration

#Include Code

Model validation

In this section we will assess how well the model performs on the training data (train split) to ensure its robustness and capabilities, so we can assess under/over fitting.

We will analyze the residuals to ensure that model assumptions are met and use R-squared as the main performance metric will the model will be computed by means of Ordinary least squares.

#Include Code

Model evaluation

Finally we will evaluate how well the model performs with unseen data (train split) to assess its forecasting capabilities, using the same R-squared performance metric.

#Include Code